

Mathematical Formulation: Plane Poiseuille Flow with Density Anomaly and Phase Change

1 System Description

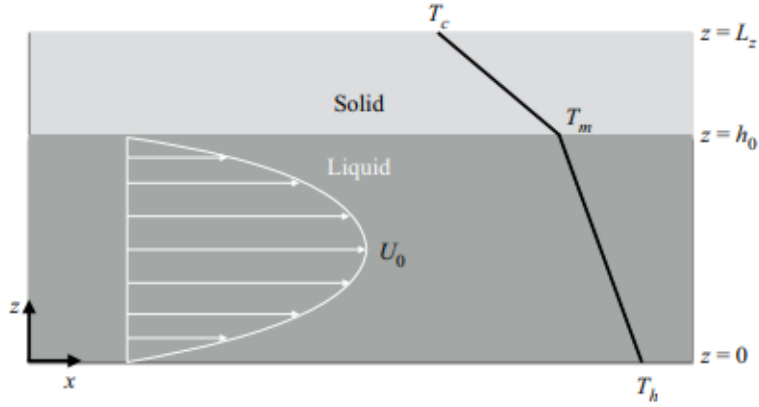


Figure 1: Schematic of the liquid-solid system with a phase-change interface.

2 Dimensional Governing Equations

For the liquid phase ($0 \leq z \leq h(x, t)$), using the Boussinesq approximation with a quadratic equation of state $\rho = \rho_{max}[1 - \gamma(T_l - T_{max})^2]$ (I took this equation from G. Veronis, Penetrative convection, Astrophys. J. 137, 641 (1963)):

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\frac{1}{\rho_{max}} \nabla p + g\gamma(T_l - T_{max})^2 \hat{\mathbf{k}} + \nu \nabla^2 \mathbf{u} \quad (2)$$

$$\frac{\partial T_l}{\partial t} + \mathbf{u} \cdot \nabla T_l = \kappa \nabla^2 T_l \quad (3)$$

For the solid ice phase ($h(x, t) \leq z \leq L_z$), :

$$\frac{\partial T_s}{\partial t} = \kappa \nabla^2 T_s \quad (4)$$

- k is the thermal conductivity
- κ is the thermal diffusivity
- ρ_{max} is the density at $T = T_{max} = 4^\circ\text{C}$
- $\gamma = 7.68 \times 10^6 / \text{C}^2$

3 Boundary Conditions

At the stationary, heated bottom wall ($z = 0$):

$$\mathbf{u} = 0, \quad T_l = T_h \quad (5)$$

At the top cold boundary ($z = L_z$):

$$T_s = T_c \quad (6)$$

At the moving ice-water interface ($z = h(x, t)$):

$$T_l = T_s = T_m \quad (\text{Phase equilibrium}) \quad (7)$$

$$\mathbf{u} \cdot \hat{\mathbf{n}} = \mathbf{u} \cdot \hat{\mathbf{t}} = 0 \quad (\text{No-slip and no-penetration}) \quad (8)$$

$$\rho_s L_s \frac{\partial h}{\partial t} = \left[k_s \frac{\partial T_s}{\partial z} - k_l \frac{\partial T_l}{\partial z} \right]_{z=h} \quad (\text{Stefan condition}) \quad (9)$$

4 Non-Dimensionalisation

We scale length by the initial channel depth h_0 , velocity by the Poiseuille centerline velocity U_0 , time by the thermal diffusion time h_0^2/κ , pressure by $\rho_{max} U_0 \kappa / h_0$, and temperature as $\theta = (T - T_m)/\Delta T$ where $\Delta T = T_h - T_m$. The non-dimensional governing equations become:

$$\nabla \cdot \mathbf{u} = 0 \quad (10)$$

$$\frac{\partial \mathbf{u}}{\partial t} + Pe(\mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \frac{Ra_\gamma Pr}{Pe}(\theta_l - \theta_{max})^2 \hat{\mathbf{k}} + Pr \nabla^2 \mathbf{u} \quad (11)$$

$$\frac{\partial \theta_l}{\partial t} + Pe(\mathbf{u} \cdot \nabla \theta_l) = \nabla^2 \theta_l \quad (12)$$

$$\frac{\partial \theta_s}{\partial t} = \nabla^2 \theta_s \quad (13)$$

4.1 Non-Dimensional Boundary Conditions

At the stationary, heated bottom wall ($z = 0$):

$$\mathbf{u} = 0, \quad \theta_l = 1 \quad (14)$$

At the top cold boundary ($z = L_z/h_0$):

$$\theta_s = -\Lambda \quad (15)$$

At the moving ice-water interface ($z = h(x, t)$):

$$\theta_l = \theta_s = 0 \quad (\text{Phase equilibrium at } T_m) \quad (16)$$

$$\mathbf{u} \cdot \hat{\mathbf{n}} = \mathbf{u} \cdot \hat{\mathbf{t}} = 0 \quad (\text{No-slip and no-penetration}) \quad (17)$$

Subject to the non-dimensional Stefan interface condition:

$$\frac{\partial h}{\partial t} = \frac{1}{\Lambda S} \left[\frac{\partial \theta_s}{\partial z} - \frac{\partial \theta_l}{\partial z} \right]_{z=h} \quad (18)$$

where the dimensionless parameters are:

$$Ra_\gamma = \frac{g\gamma(\Delta T)^2 h_0^3}{\nu \kappa}, \quad Pe = \frac{U_0 h_0}{\kappa}, \quad Pr = \frac{\nu}{\kappa} S = \frac{L_s}{c_p(T_m - T_c)}, \quad \Lambda = \frac{T_m - T_c}{\Delta T}, \quad \theta_{max} = \frac{T_{max} - T_m}{\Delta T} \quad (19)$$

5 Base State

The steady, fully developed base state is: We assume a steady ($\partial/\partial t = 0$), fully developed, unidirectional base flow ($\bar{w} = 0$, $\partial/\partial x = 0$) with a planar, stationary ice-water interface at $z = 1$. Under these assumptions, the governing equations reduce to 1D boundary value problems. The base state profiles are:

$$\bar{u}(z)^{(0)} = 4z(1 - z) \quad \text{(Plane Poiseuille flow),} \quad (20)$$

$$\bar{\theta}_l^{(0)}(z) = 1 - z \quad \text{(Linear conduction in liquid),} \quad (21)$$

$$\bar{\theta}_s^{(0)}(z) = \frac{\Lambda}{d_0}(1 - z) \quad \text{(Linear conduction in solid)} \quad (22)$$

$$\bar{d}_0 = \Lambda \quad \text{From Stefan condition} \quad (23)$$

$$(24)$$

where $d_0 = L_z/h_0 - 1$ is the non-dimensional ice thickness. Substituting these thermal profiles into the Stefan condition yields the heat flux balance required to maintain a stationary interface.